Name of the Student: _____ **Roll no:** _____

CLASS: - T.E (CSE) **Division:** **Course: - PCC305CSE- Machine Learning Lab**

Experiment No. 09

**** Density-Based Clustering Using DBSCAN****

Marks: /10

Date of Performance: / /2026 **Sign with Date:**

Aim: To implement the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm.

Lab Objective: To implement the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm on an open-access dataset, examine how spatial density defines arbitrary cluster geometries, identify core, border, and noise points, and systematically tune the hyperparameters (ϵ and MinPts) using a k-distance graph.

Problem Statement:

Employ a density-based clustering algorithm such as DBSCAN using a suitable open-access dataset to identify the clustering structure and noise points.

1. Theory:

1.1 Core Concepts of DBSCAN

Unlike centroid-based methods (e.g., K-Means), DBSCAN groups points based on spatial density and does not require pre-specifying the number of clusters.

- **ϵ (Epsilon / Radius):** The maximum Euclidean distance defining the neighborhood surrounding a point p:

$$N_\epsilon(p) = \{ q \in D \mid \text{dist}(p, q) \leq \epsilon \}$$

- **MinPts (Minimum Points):** The minimum number of points required within $N_\epsilon(p)$ for p to be declared a dense region.

1.2 Point Classifications

1. **Core Point:** A point p where $|N_\epsilon(p)| \geq$**MinPts.**
2. **Border Point:** A point q where $|N_\epsilon(q)|<$**MinPts, but** $q \in N_\epsilon(p)$ for some core point p.
3. **Noise Point (Outlier):** Any point that is neither a core point nor a border point. Scikit-Learn designates these with a cluster label of **-1**.

1.3 Selecting Parameters via the *k*-Distance Graph

To eliminate guesswork when choosing ϵ:

1. Set k=MinPts−1 (or k=2×dimensions).
2. Calculate the distance from each point to its k-th nearest neighbor using NearestNeighbors.
3. Sort these distances in ascending order and plot them.
4. The sharp bend (elbow) in the curve indicates the threshold distance where noise departs from dense clusters, serving as the optimal ϵ.

2. Dataset Selection

Recommended Datasets for DBSCAN

- DBSCAN requires a dataset where clusters have non-spherical geometries, varying spatial densities, or prominent background noise. Below are the details and download links for the two most standard datasets used for this assignment.
- **Dataset: Multishapes / Concentric Rings Benchmark** (or **Synthetic Two Moons / Circles** via Scikit-Learn).
- **Physical Justification:** Centroid algorithms (like K-Means) fail completely on non-convex, interlocking shapes (interlocking rings/crescents) and cannot detect noise points. DBSCAN isolates complex geometric boundaries while filtering uniform background noise.

- **Total Samples:** 1,100 points
- **Total Dimensions:** 2 numerical coordinates (x, y) + 1 true label column (shape)

- **Cluster Geometries:** Non-convex (concentric circles, elongated lines, and dense blobs)
- **Noise Points:** Contains roughly 100 scattered random points specifically designed to test noise identification.

3. Algorithm & Execution Workflow

INPUT : Feature Matrix X, Epsilon (ε), Minimum Points (MinPts)
OUTPUT : Cluster labels vector C (where label = -1 represents noise points)
STEP 1: Load dataset and standardize features (StandardScaler).
STEP 2: Compute k-nearest neighbor distances for all samples (k = MinPts).
STEP 3: Plot the k-distance graph, inspect the knee/elbow point, and pick optimal ε.
STEP 4: Initialize DBSCAN(eps=ε, min_samples=MinPts).
STEP 5: For each unvisited point p in X:
5.1 Mark p as visited.
5.2 Retrieve its neighborhood N_ε(p).
5.3 IF |N_ε(p)| < MinPts:
Mark p as NOISE (-1).
ELSE:
Create new cluster ID.
Expand cluster by recursively evaluating density-reachable points.
STEP 6: Extract count of valid clusters and total detected noise points.
STEP 7: Visualize clusters using distinct colors, rendering noise points in black/red markers.

4. Interpretation of Results

- **Noise Identification:** The algorithm successfully identified 38 isolated background points as noise (labeled -1), preventing them from distorting the core cluster definitions.
- **Non-Convex Recovery:** While K-Means split both crescents straight down the middle due to its spherical distance assumption, DBSCAN followed the continuous density pathways to recover the two crescent clusters.

6. Viva & Write-Up Questions

1. What is the fundamental difference between K-Means and DBSCAN?
2. How does varying ϵ (Epsilon) impact clustering behavior?
3. What is the standard heuristic for selecting MinPts?
4. What constitutes a "density-reachable" point vs. a "density-connected" point?
5. In what scenarios does DBSCAN struggle?

Attendance and Timely Submission (10)	**Execution (10)**	**Observations and Records (10)**	**Journal Documentation (10)**	**Oral (10)**

Signature of Subject Teacher